OPN (MAIN PLANT) Department
SIMHADRI

Date: 12/05/2022(16:34:37)

Ref:SMTPP/OPN (MAIN PL/2022-23/OPRN/96602

Subject:REVISION:3 LMI ON OPTIMISATION OF WATER CONSUMPTION IN ASH HANDLING SYSTEMS

LMI/OGN/OPS/SYST/004 Issue No.: 1 Rev: 03 Date: May 2022"OPTIMISATION OF WATER CONSUMPTION IN ASH HANDLING SYSTEMS" has been reviewed as periodical review.No changes made in this revision.

Put up for kind approval.

Submitted by:

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May please be approved.

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May pl be approved

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Unauthorized Use Prohibited

Submitted for approval of competent authority

Submitted by:

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Pl review

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Reviewed and put up for approval please.

Submitted by:

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Dry ash running hours is made available for all passes of St-1 and St-2 Units in the AHP Overview accessible from Intranet and also through PI Vision (Intranet through login) and can be monitored for complying to Clause 9.20 of the LMI.

Forwarded for further review

Submitted by:

टी.आर. राजगोपालन् - अपर महाप्रबंधक

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May please be approved.

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Reviewed

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LMI reviewed, and the revision (no changes envisaged) may please accorded approval by HOP, Simhadri.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

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कृपया अनुमोदन प्रदान किया जाए।

May be approved please.

Submitted by:

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NTPC Ltd.

Simhadri Thermal Power Station

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/SYST/004

Issue No.: 1 Rev: 03   Date: May 2022

OPTIMISATION OF WATER CONSUMPTION IN ASH HANDLING SYSTEMS

Approved for
Implementation by ....
HOP (Simhadri)

Date: ....

O & M OPERATIN DEPARTMENT

Ref: SMPP/OPRN/LMI/ OGN/OPS/SYST/00402 **Date: 12.05.2022**

The following LMI has been revised as per periodical review of 2 years period.

Sl.No	LMI revision details	LMI (Name)	Reason for revision	LMI Revision Dt.
1	LMI/ OGN/OPS/SYST/004 Issue no: 1 Rev no: 3, Dt: May'2022	Optimization of Water Consumption In Ash Handling Systems	Periodical review	12.05.2022

Put up for kind approval please

	Name	Designation	Signature
Prepared by:	**G VENKATA SUBBAIAH**	**Sr. Manager (O)**	
Checked by:	**DR.TV RAJASEKHAR**	**AGM(O)**	
Checked by:	**PALA RAJASEKHAR**	**AGM(O)**	
Checked by:	**NILANJAN MANDAL**	**AGM(AHD)**	
Checked by:	**T.R. RAJAGOPALAN**	**AGM(C&I)**	
Re-Checked by:	**MATHEW EIPE KOVOOR**	**GM(Maintenance)**	
Reviewed by:	**RK JHA**	**GM (O&M)**	
Approved by	**GIRISH CHANDRA CHOUKSE**	**GGM (Simhadri)**	

INDEX

1.0	**INTRODUCTION**
	At SMTPP, The bottom ash from boiler is evacuated by employing water impounded jet pump system. The fly ash from ESP hoppers (which constitutes 80% of total ash generated) is also normally evacuated by using pressure conveying (dry) method / wet collection method. The ash handling system is mainly classified into a) Ash water system b) Bottom ash system c) Fly ash (Wet/Dry) system d) Ash disposal system e) Ash water recirculation system. All the above systems except dry ash system requires mixing of water with ash for conveying it to disposal area. Sea water is being used for ash handling purpose.
	Ash water system:-
	This system provides the necessary water required by the ash handling system in bottom ash as well as fly ash evacuation & disposal system. Water from ash water recirculation system and from CW blow down are used as the source of make up to this system.
	Bottom Ash System:-
	Bottom ash, resulting from combustion of coal inside the boiler, is continuously collected in water impounded bottom ash hopper for its periodic removal. Bottom ash gets cooled as it enters into the water thus avoids the clinker formation. The mixture of ash and water is discharged through feed gate to clinker grinder which grinds the clinkers to smaller size so that it can be conveyed through pipe line. Crushed clinkers in slurry form falls in to the hydro ejector. Hydro ejector provides jetting action by means of high pressure water flowing through it to convey the slurry through pipe lines and various valves to the common Ash Slurry Sump. The Bottom ash hopper over flow is collected in bottom ash over flow tank from there it can be pumped to ash slurry sump.
	Fly ash system:-
	Fly ash collected from electrostatic precipitator collection hoppers is continuously removed by vacuum extraction system to collector tank in wet mode type of evacuation system. The vacuum system is used to convey the fly ash from their respective collection point to collector tank, where it is mixed with water through wetting head. Fly ash from all passes of ESP Hoppers can be collected in Buffer Hopper in the dry form from where it is transported to main silo. In the collector tank fly ash conveyed through vacuum line is mixed with water and the resultant slurry is transported to **ash slurry sump** through pipe line in stage-1 and through open trench by gravity in stage-2 units. The ash slurry sump is common for **bottom ash** collection and **fly ash** (wet system) collection. The **FG duct hoppers** at Economizer outlet duct (4 nos of each unit) and APH hoppers (8 nos for each unit) are evacuated through the flushing apparatus using FAHP water in stage-II units and in stage-I units all APH hoppers connected to Fly ash system E2 pass & all FG duct hoppers at Economizer outlet duct (4 nos of each unit) discharge is connected to the Bottom ash hopper.
	Ash Disposal system:-
	Both bottom ash & fly ash collected in common ash slurry sump are disposed through ash disposal pump series **(5 nos in stage-I and 4 nos in stage-II)** to the ash dyke located at about

	6 KM distance from pump house. Each series is provided with **2 pumps** in series and 1st pump being controlled by hydraulic scoop.
	Ash Water Recirculation system :-
	The function of ash water recirculation system is to pump decanted ash water from ash pond clean water lagoon to the plant with the help of ***ash water recirculation pumps*** installed in ash water recirculation pump houses of Stage-I & stage-II. Ash water recirculation water is directly fed into the **ash water tank** for use in ash handling system or it can be pumped to EDP sump and further back to sea through EDP pumps as per the system requirement.
2.0	**SUPERSEDED DOCUMENTS & REVISION NOTES**
	Superseded: LMI/OGN/OPS/SYST/004; Issue1/ Rev-2, Dt: May'2022 Rev01 Jan'2020 - Revision notes introduced in paragraph 2.0. Rev02 May 2020 - 1) Addition of Clause 8.1 on Ash Disposal pumps power saving 2) Addition of Clause 8.3 on ash water recirculation - Recycling 3) Addition of sub clause 9.12 re-use of decanted water from ash dyke 4) Checklist for Water optimization added as Annexure1 Rev03 May 2022-NIL
3.0	**SCOPE**
	Scope of this LMI document is 4 X 500 MW Units of Simhadri STPS, Stage-I & II.
4.0	**REFERENCES :-**
	1. OGN: COS-ISO-00-OGN/OPS/SYST/004, Rev. No.: 1, Date: April 2020 2. Ash handling system documents supplied by M/S Mahendra Ashtec for stage-1 units and M/S Indure systems for stage-2 units.
5.0	**NEED FOR OPTIMISATION OF WATER CONSUMPTION**
5.1	The concentrations at which bottom ash and fly ash are being transported vary between 10 to 30% by weight. This can become uneconomical due to increase in the cost of hydraulic transportation. Hence consumption of water in ash slurry has direct impact on pump power, therefore, optimum use of water will definitely reduce the O&M cost.
6.0	**METHODS FOR MINIMISING WATER CONSUMPTION**
	During evacuation and disposal of bottom / fly ash, excess quantity of water is used in making ash slurry thus exceeding the designed value of ash water ratio. Efforts must be made to optimize water consumption as far as possible. Even 10-15% reduction in water consumption will save lot of money by stopping either of ash water pumps or ash slurry pumping series. The sources of excess water consumption and method to control water consumption are described as below:
6.1	Optimization / control of ash water by throttling valves at individual HP / LP pump discharge or at common discharge header.

	Responsibility-AHP Engr.
6.2	Optimization / control of ash water by throttling valves at individual utility point (Wetting Unit, FAHP to Eco Hoppers, Manual Ejectors, BA flushing, filling, Grinder Seal Water, BA Filling, Refractory & Seal trough, Vacuum pump, Grinders & BAOF Seal water). **Responsibility**-UCE/AHP Engr.
6.3	Optimization of water consumption in bottom ash seal trough area by Controlling of water flow through make up / overflow pipe lines to BAOF tank. **Responsibility**-UCE
6.4	Reducing size of flushing nozzles in ESP wetting units/APH/ECO flushing apparatus (unit-1&2 Economizer hoppers and unit-3&4 APH and Economizer hoppers) **Responsibility**-HOD(AHD)
6.5	Blanking of HP/LP water nozzles in ESP trench / Fly ash common trench / Fly ash sump wherever possible. (U4 C & D PASS, Unit1&2 ESP new trench) Trench nozzles can be blanked depending upon the slope of trench available for smooth flow of fly ash slurry. **Responsibility**-HOD(AHD)
6.6	Operation of intermittent fly ash evacuation system (wherever provided) as per designed mode -this system is designed to store ash for a specified period. However site O&M operate the system on continuous basis .If the system is run intermittently O&M flexibility will increase and lot of pumping power of ash slurry disposal pumps will reduce ,thus minimizing excess pumping of water. Depending upon coal flow and hopper levels it can be followed. **Responsibility**-AHP Engr.
6.7	Control of makeup water in bottom ash and fly ash slurry sumps keeping in view the specified values of slurry concentrations and minimum velocity for slurry transportation. The maximum velocities in slurry transportation for both Fly ash and Bottom ash should be limited to 2.8 meter per second. The minimum velocity should be above the critical settling velocity, which depends upon ash particle size. **Responsibility**-AHP Engr.
6.8	The designed maximum concentration of B.A. and F.A. slurries when being handled independently or combined should be as under: a) Bottom Ash (B.A.) - 25% (w/w) b) Fly Ash (F.A.) - 30% (w/w) c) Combined B.A. & F.A. - 25% (w/w) **Responsibility**- AHP Engr./HOD(AHP)
7.0	**MEASUREMENT OF ASH WATER CONSUMPTION**
7.1	The measurement of water consumed for making ash slurry is ash / water ratio by weight / weight (w/w). To increase awareness among O&M executives for water consumption weekly monitoring of ash water ratio should be made mandatory. Ash water sample can be collected from common slurry trench if possible otherwise the same can be collected from discharge point at ash bund. As per design, stage1 Ash water ratio is 7.68 & stage2 ash water ratio 7.43 approx.

	Responsibility :HOD(Chem)
8.0	**SAVING OF PUMPING POWER**
8.1	**Slurry Disposal Pumps**
	By minimizing water consumption in making ash slurry, one pumping series can be stopped for certain duration by operating the pump series with increased speed. The first stage of slurry disposal pumps are provided with variable speed hydraulic coupling for setting the optimum operating speed of the pumps based on actual site requirements. Minimum (+) 10% and maximum (+) 75% speed variation with respect to speed at rated design point is provided in the first stage of combined ash slurry disposal pumps . Variation of +10% in pump speed can also be achieved in subsequent pump stages (where belt drive arrangement is provided) by replacing the existing belt pulley with a pulley of different diameter. Continuous motor rating (nameplate rating) at 50^{0}C for ash slurry pumps have at least 20% margin above the maximum load demand of driven equipment in the entire operating range.
	Responsibility-AGM(O)/AGM(AHD)
8.2	**Ash Water Pumps (HP/LP Pumps)**
	Suitable number of high pressure and low pressure pumps (with standby facility) are provided for ash slurry extraction and disposal. With the reduction in water consumption in ash handling system, one or two HP/LP pumps can be stopped (continuous or intermittently) thus saving in auxiliary power consumption. This condition can be achieved more effectively by arresting all the leakages (in valve glands and pipings) in the system and with proper control of slurry concentration. The operating engineers have to play a vital role in implementing the above operating modes.
	Responsibility- AGM(O)/AGM(AHD)
8.3	**Ash Water Recirculation from Ash Dyke:**
	Water Sent to ash dyke along with slurry is to be recycled back to Ash water sump in side plant premises through AWRS system. About 80% of the water can be easily reclaimed through our system.
	Responsibility-AGM(O)/AGM(AHD)
9.0	**OPERATION AND MAINTENANCE PRACTICES**
	General :-
9.1	To increase awareness among Operation executives about water conservation, aspects of water conservation are to be discussed in communication meeting.
	Responsibility-SCE
9.2	The ash water ratio is to be measured on weekly basis by chemistry dept and to be put on LAN and Water balance study to be done.
	Responsibility-HOD(Chemistry) / HOD(EEMG)
9.3	Ash water pipe line and valve gland leakages in the ash water system are to be arrested immediately without waiting for unit shutdowns wherever it is possible. With the reduction in water consumption in ash handling system, optimization of HP/LP pumps running hours can be done for saving in auxiliary power consumption.
	Responsibility-AGM(AHD)/AHP Engr

9.4	Optimization of ash water make up to ash slurry sumps and ash water sumps to be done by remote control valves and avoid overflow.
	Responsibility-AHP Engr
9.5	Optimization of ash water by closing standby collecting tower wetting unit water valves.
	Responsibility- AHP Engr
9.6	Bottom ash hopper refractory cooling water and Seal trough makeup to be optimized to avoid excessive overflowing of water. The Refractory cooling water and seal trough makeup water to be isolated during Unit shutdown period to reduce unwanted over flow from BA Hopper**.**
	Responsibility- SCE / UCE
9.7	Optimization of flushing nozzles size in Wetting unit/APH / ECO hoppers flushing apparatus (unit-1&2 Economizer hoppers and unit-3&4 APH and Economizer hoppers)
	Responsibility- AGM(O)/AGM(AHM)
9.8	Blanking of HP water nozzles in ESP trench (U4 C & D PASS, Unit1&2 ESP new trench) / Fly ash common trench / Fly ash sump wherever possible. Trench nozzles can be blanked depending upon the slope of trench available for smooth flow of fly ash slurry.
	Responsibility- AGM(O)/AGM(AHM)
9.9	Control of makeup water in bottom ash and fly ash slurry sumps keeping in view the specified values of slurry concentrations and minimum velocity for slurry transportation.
	Responsibility- AGM(O)/AGM(AHM)
9.10	Optimization of ash slurry series pumps running hours when sump level is permitted.
	Responsibility- SCE / AHP Engr
9.11	The following tanks are to be prevented from overflowing i. Bottom ash over flow tank ii. Ash water tank **Responsibility**- UCE/ AHP Engr
9.12	All seepage / decanted water/ toe drain is required to be collected at NTPC land. The collected water is to be recycled back for suppressing of fugitive dust emission in ash dyke.
	Responsibility- AGM(AHD)&AGM(O)
	Bottom Ash handling
9.13	Periodic replacement of hopper flushing nozzles and adjustment of nozzle fixing angle to be done as per the requirement irrespective of unit shutdowns.
	Responsibility- AGM(AHD)
9.14	Use of fire water/service water on continuous basis in ash handling system shall be avoided. Only to be used in case of emergency. **Responsibility**- AGM(O)/AGM(AHD)
9.15	In case of unit shutdown, Hydro ejector water valves are to be kept closed when there is no ash evacuation. If any valve passing is there, then it is to be brought to the notice of AGM (AHD).
	Responsibility- SCE / AGM(O)/AGM(AHD)
9.16	Seal trough and BAH make flow to be regulated
	Responsibility- UCE

9.17	Passing of BA Hopper makeup valve / Hopper Jetting Valves / Hydro-ejector valves are to be closely monitored and attended as any valve passing will lead to more consumption.
	Responsibility- UCE/AGM(AHD)
9.18	Effort shall be put to maintain BAHP pressure by maintaining the pump internals / Valves so that proper de-ashing is carried out by operating one BAHP pump each for U 1&2, U 3&4 respectively.
	Responsibility- AGM(O)/AGM(AHD)
9.19	Passing of bottom ash hopper gates and BA discharge valves are to be monitored and problems to be rectified at the earliest opportunity. **Responsibility**- UCE/AGM(AHD)
	Fly Ash Handling
9.20	Preference to be given for dry ash disposal for ease in disposing ash from ESP hopper. Record of Dry ash and wet ash disposal running hrs to be maintained. Issues in dry ash disposal are to be resolved at earliest to shift from wet ash to dry ash disposal.
	Responsibility- AGM(O)/AGM(C&I)/AGM(AHD)
9.21	The water to flushing apparatus provided in unit-1&2 Economizer hoppers and in unit-3&4 APH and Economizer hoppers are to be kept closed when unit under is shutdown & hoppers got empty.
	Responsibility- UCE
9.22	HP water to be used for flushing apparatus as per requirement and corrective action to be taken in case of fall of pressure.
	Responsibility- AGM(O)/AGM(AHD)
9.23	Wetting head water pneumatic isolation valves should work from remote and ensure perfect isolation of water supply to the **CT**. Whenever any **CT** is not in service the manual isolation valve needs to be closed. **Responsibility**- AHP Engr/AGM(C&I)/AGM(AHD)
9.24	Stand by Vacuum Pump Cooling/Sealing water to be kept isolated.
	Responsibility- SCE/AHP Engr
9.25	The fly-ash sluice trench cleanliness is to be maintained & silting of ash in deep trench is to be avoided to exploit the trench capacity. This will avoid the starting of additional series for high sump level during short duration. The fly-ash sluice trench cleaning is to be carried out during annual overhaul.
	Responsibility- AGM(O)/AGM(AHD)
9.26	The CW blow down valves and Ash water recirculation make up valve to Ash water tank should work on level interlock and thus prevent over flow of sump. These valves are to be maintained in 'Auto' mode during unit operation and any passing / non-functioning of these valves to be attended on priority.
	Responsibility- AGM(O)/AGM(C&I)/AGM(AHD)
10.0	**RECORD : -**
10.1	The daily activities of ash handling system are recorded in stage-1 & stage-2 ASPH Engineers log Book / UCE log book / SCE log books. (BA duration, FA system operation, Running

	Hours Etc).
	Responsibility- AHP Engrs/UCE
11.0	**REVIEW : -**
11.1	The application & implementation of this LMI shall be reviewed by Head of Operation once in **3** years.
12.0	**Distribution list**
	Simhadri Local Area Network

ANNEXURE1

S.No	Check Point For Water Optimization	Responsibility	Status / Remarks
1)	Measurement of Ash Water Ratio by weight on weekly basis & uploading in intranet for awareness and corrective measures. (Design Values = St1 -7.68; St2- 7.43)	Chemistry	
2)	Optimization of Series pumps running hours	Operation	
3)	Optimization of HP/LP pumps running hours	Operation	
4)	Optimization of series pumps scoop and discharge pressure.	Operation	
5)	Maximizing Dry ash running hrs and reducing wet ash disposal hours.	Operation/AHD	
6)	Optimization of ash water make up to ash slurry sumps (Avoiding makeup to just run additional series & avoiding passing of makeup valves)	Operation/AHD	
7)	Optimization of water to ash water sumps to be done by auto operation of valves and avoiding overflow & passing of valves.	Operation/AHD/ C&I	
8)	Avoiding overflow of Ash sumps and BAOF sumps	Operation	
9)	Unit shutdown period – Stopping of Refractory cooling, seal trough makeup water. Closing of hydro ejector water valves, APH Eco Hopper flushing valves when there is no ash evacuation.	Operation	
10)	Optimization of ash water by closing standby collecting tower wetting unit water valves & standby vacuum pumps.	Operation	
11)	Attending of passing of BA Hopper makeup valve /Hopper Jet Valves/ bottom ash hopper gates / BA discharge valves	Operation	
12)	Optimization ash water by throttling valves at individual utility points (Wetting Unit, FAHP to Eco Hoppers,	Operation	

	Manual Ejectors, BA flushing, filling, Grinder Seal Water, BA Filling, Refractory & Seal trough, Vacuum pump, Grinders & BAOF Seal water) while avoiding passing of valves		
13)	Optimization of flushing nozzles size in ESP wetting units/ APH / ECO hoppers flushing apparatus (unit-1&2 Economizer hoppers and unit-3&4 APH and Economizer hoppers).	AHD	
14)	Blanking of HP/LP water nozzles in ESP trench / Fly ash common trench / Fly ash sump wherever possible. (U4 C & D Pass, Unit1&2 ESP new trench)	AHD	
15)	Periodic replacement of hopper flushing nozzles and adjustment of nozzle fixing angle	AHD	
16)	AWRS water maximum utilization for Makeup to ash sump while reducing blowdown.	Operation	